Reading and Comparing Multi-Digit Whole Numbers Answer Key

Grades 4–5 Number and Operations in Base Ten

Quick Answers

1. 47382, 7000	5. 209040	9. 40000
2. 10	6. 8921, 80912, 80921	10. 35000, —
3. 538062	7. 47000	
4. —	8. 50000	

Curriculum

Level: Grades 4–5 Number and Operations in Base Ten
2.NBT.A – *problems 1, 2, 3, 4, 5, 6, 8, 10*
4.NBT.B.4 – *problems 7, 9, 10*
Difficulty 1–4 of 5 across 12 tagged checks.

1. Place-value chart: 4 ten thousands, 7 thousands, 3 hundreds, 8 tens, 2 ones.
(a) Multiply each digit by the value of its place and add: $4 \times 10{,}000 = 40{,}000$; $7 \times 1{,}000 = 7{,}000$; $3 \times 100 = 300$; $8 \times 10 = 80$; $2 \times 1 = 2$. Adding gives the standard form. $\boxed{47{,}382}$
(b) The 7 sits in the thousands column, so it is worth 7 thousands, not 7 ones. $\boxed{7{,}000}$

2. Each place to the left is worth ten times the place to its right. The left 4 is worth 40,000 and the next 4 is worth 4,000, so divide: $40{,}000 \div 4{,}000$. $\boxed{10}$
Say it as a sentence: "the left 4 is worth ten times as much as the 4 beside it."

3. Each addend names one place, so read the digits straight off: 5 hundred thousands, 3 ten thousands, 8 thousands, **0 hundreds**, 6 tens, 2 ones. The hundreds place has no addend, so a zero must hold that place. $\boxed{538{,}062}$
Watch the zero: writing 53,862 drops the hundreds place and shrinks the number by a factor of ten.

4. Compare from the left. Ten thousands: $4 = 4$. Thousands: $6 = 6$. Hundreds: $3 = 3$. Tens: 1 versus 5 — this is the first place where the digits differ, and $1 < 5$, so the first number is smaller. $\boxed{46{,}315 < 46{,}351}$
The ones digits (5 and 1) never get compared: a difference found in a higher place decides the comparison.

5. "Two hundred nine thousand" is the thousands period: 209 thousands, so $200{,}000 + 9{,}000$. "Forty" is 4 tens. There are no hundreds and no ones, so zeros hold those places. $\boxed{209{,}040}$

6. First count digits: 8,921 has four digits, the other two have five, so it is the smallest. Then compare 80,912 with 80,921: they agree in the ten thousands, thousands and hundreds places; in the tens place $1 < 2$, so 80,912 comes first. $\boxed{8{,}921 < 80{,}912 < 80{,}921}$

7. To round to the nearest thousand, look at the hundreds digit — it tells you which side of the halfway point you are on. In 47,382 the hundreds digit is 3, so the number is below the halfway point 47,500 and rounds down to the thousand below. $\boxed{47{,}000}$

8. In 350,842 the digits are 3 hundred thousands, 5 ten thousands, 0 thousands, 8 hundreds, 4 tens, 2 ones. The 5 sits in the **ten thousands** place, so its value is $5 \times 10{,}000$. $\boxed{50{,}000}$
Mia used the hundred thousands place, which belongs to the 3.

9. Round each number to the nearest thousand, then add the rounded numbers. 18,472: hundreds digit 4, so round down to 18,000. 21,650: hundreds digit 6, so round up to 22,000. Then $18{,}000 + 22{,}000$. $\boxed{40{,}000}$
The exact total is 40,122 books, so the estimate is close — that is the point of estimating: a quick answer you can trust to the nearest thousand.

10. **(a)** 34,650: the hundreds digit is 6, which is above the halfway point 34,500, so round up to 35,000. (Belmont's 34,509 also rounds to 35,000.) $\boxed{35{,}000}$
(b) Rounding threw away the information that separates the two towns, so compare the original numbers place by place: ten thousands $3 = 3$, thousands $4 = 4$, and then the hundreds place, 6 against 5. Since $6 > 5$, Ashford has more people. $\boxed{34{,}650 > 34{,}509}$
Two numbers can round to the same thousand and still be 141 apart — an estimate is never a substitute for a comparison.